Solar Shading — User Guide (Revit 2026)

A Revit add-in that computes the **shaded area of windows from external shading devices**, the resulting **external shading coefficient (SC2)**, and the **Singapore BCA ETTV** with pass/fail. It also casts a building's shadow onto the ground.

> The sun position is computed analytically from the project's site location — no Revit sun-study transaction is needed, so the analysis is fast and does not depend on the active view's sun settings.

---

1. Install

From the release package (recommended). Download `SolarShading-<version>.zip` from the [Releases page](), then with **Revit closed**:

1. Unzip it anywhere.
2. Right-click `Install.ps1` → **Run with PowerShell** (or run `.\Install.ps1`).

It detects every Revit 2025–2027 on the machine and installs the matching build (.NET 8 for 2025/2026, .NET 10 for 2027) into `%AppData%\Autodesk\Revit\Addins\<version>\`.

```
.\Install.ps1 -RevitVersion 2026   # a single version
.\Install.ps1 -AllUsers            # machine-wide (run PowerShell as Administrator)
.\Uninstall.ps1                    # remove the add-in
```

If Windows blocks the script, run `Set-ExecutionPolicy -Scope Process -Bypass` once in that PowerShell window and try again.

Building from source instead? With Revit closed, run `./deploy/Deploy.ps1 -RevitVersion 2026` from the repository — it builds and installs in one step.

First launch: Revit shows "*The publisher of this add-in could not be verified*" because the DLL is unsigned. Click **Always Load**. A **Solar Shading** ribbon tab will appear.

---

2. Before you run — model checklist

1. **Site location is set** — *Manage → Location →* set Latitude/Longitude (and time zone). The sun position depends entirely on this. The default Boston location gives Boston sun.
2. **Project North** is set correctly if your model isn't aligned to true north.
3. **Windows are hosted in walls** (standard window families). Curtain walls are partially supported.
4. **Shading devices have 3D solid geometry** (overhangs, fins, ledges, projecting slabs, mullions…).
5. *(Optional)* Windows carry **analytic thermal properties** (U-value, SHGC) on the family/type so ETTV uses the real glass. Missing values fall back to the glazing you pick in the dialog.

---

3. Workflow

Step 1 — Tag the shading devices

1. Select the elements that shade the windows (overhangs, fins, ledges, projecting floors…).
2. **Solar Shading → Get Shading Devices**. This sets a `SS_SHADING_DEVICE` = Yes flag on the selection. Re-run any time to add more.

Step 2 — Run the shading + ETTV analysis

1. **Solar Shading → Shading on Windows**.
2. In the dialog choose:
3. **Reference dates** — 21 Mar / 21 Jun / 21 Dec (equinox / solstices).
4. **Time of day** — hour range (default 7–18) and the year.
5. **Glazing (fallback)** — used only for windows with no thermal data in the model.
6. **Wall U** and **ETTV threshold** (default 50 W/m²).
7. **Output** — show shadow overlay, write to shared parameters, export CSV.
8. Click **Run**. The **ETTV Results** window shows a per-orientation table (Window area, WWR %, SC2, ETTV) and the envelope **ETTV — PASS / FAIL**.

Step 3 — Building shadow on ground (optional)

1. Select one or more **Mass** elements.
2. **Solar Shading → Building Shadow on Ground**, then pick a **date and hour**. Draws the cast shadow as a coloured DirectShape on the ground and reports its area. Re-running replaces the previous building-shadow overlay.

4. Where the results go

Output	Location
External shading coefficient	Window shared parameter `SS_EXTERNAL_SC2`
Hourly shaded area (m^2) per date	`SS_SHADED_MARCH` / `SS_SHADED_JUNE` / `SS_SHADED_DEC` (semicolon-separated, one value per hour, `NA` when the window doesn't face the sun)
Full hourly table	**CSV on your Desktop** — `SolarShading_YYYYMMDD_HHmmss.csv`
Printable compliance report	**HTML on your Desktop** — `SolarShading_Report_YYYYMMDD_HHmmss.html` (opens automatically when enabled)
Envelope ETTV + pass/fail	**ETTV Results** window
Shadow preview	Red **overlay** on each window (re-drawn each run, never stacked)

You can schedule the `SS_*` parameters in Revit (they are bound to the Windows category).

5. Interpreting the numbers

- **SC2** (0–1): the share of solar gain that still reaches the glass after the external shading — solar-radiation-weighted across the analysed hours. Lower = more effective shading.
- **ETTV** (W/m^2): `12·(1−WWR)·Uw + 3.4·WWR·Uf + 211·WWR·CF·SC1·SC2`, area-weighted across façades. **≤ 50 W/m^2 passes** (BCA Green Mark default).
- **Shaded area series**: hour-by-hour shaded area on the glass — useful for transient studies.

6. Troubleshooting

Message	Cause / fix
"No windows found"	Model has no window family instances.

Message	Cause / fix
"No elements are flagged as shading devices"	Run **Get Shading Devices** on the shades first.
"No window received shading…"	Shades are >4 m from any window, or below/behind them.
Sun below horizon (ground shadow)	The site/time gives night; check *Manage → Location*.
Results look wrong by a fixed rotation	Check **Project North** and the site location angle.

---

7. Important limitations (read before using for submission)

- This is a **geometric** shading tool (correct sun position + exact shadow areas). It is **not** a validated daylight/energy engine (Radiance-class) — use it for ETTV/shading, not for sDA/ASE.
- **Regulatory constants are indicative** and must be verified against the current code edition: the BCA solar correction factors, the 50 W/m² threshold, and the built-in glazing library values.
- **SC2 weighting** uses an ASHRAE clear-sky proxy, not a project EPW weather file.
- The window outline uses the rough opening when available, else the largest glazing face.
- Always confirm a few results by hand or against a reference tool before relying on them.